

Revenue & Business Growth Analysis

Olist E-Commerce Dataset

Business Intelligence & Data Analysis Report

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Business Question:

Revenue & Business Growth

Tools Used:

Python (Pandas, NumPy, Matplotlib, Seaborn)

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1. Introduction

Revenue growth is one of the most important indicators of business success in e-commerce. Analyzing sales performance helps organizations understand customer purchasing behavior, evaluate business growth, and identify opportunities to improve profitability.

This report analyzes Olist's Revenue and Business Growth using transactional data from the Olist e-commerce dataset. The analysis focuses on Gross Merchandise Value (GMV), revenue trends, Average Order Value (AOV), order volume growth, product category performance, and the impact of freight costs on business performance. The findings provide business insights and recommendations to support strategic decision-making and sustainable business growth.

2. Business Question

How has Olist's revenue grown over time, and what factors influence business growth, customer spending, and overall sales performance?

3. Data Preparation

3.1 Data Cleaning

The analysis was performed using the Olist e-commerce datasets, including orders, order items, products, and payments.

Before conducting the analysis, duplicate records were removed where necessary, missing values were examined, and date columns were converted to the datetime format to ensure accurate time-based analysis. The relevant datasets were then merged to create a unified dataset for revenue analysis.

3.2 Feature Engineering

Several business indicators were created to evaluate revenue performance:

- Gross Merchandise Value (GMV)
- Average Order Value (AOV)
- Monthly Revenue
- Monthly Order Volume
- Freight Cost
- Revenue Growth Rate

These indicators were used throughout the analysis to evaluate business growth, customer purchasing behavior, and overall financial performance.

4. Analysis and Results

4.1 Gross Merchandise Value (GMV)

Results

Gross Merchandise Value (GMV) was calculated as the total value of all products sold through the Olist platform during the analysis period.

The analysis showed that the total **Gross Merchandise Value (GMV)** reached:

GMV = 13,591,643.70 BRL

This value represents the total sales generated before considering shipping costs, discounts, or other operational expenses.

Business Insight

The high GMV indicates that Olist generated substantial sales during the analyzed period, reflecting strong marketplace activity and customer demand. GMV serves as a key business performance indicator and provides a baseline for evaluating revenue growth, product category performance, and customer purchasing behavior in the following analyses.

4.2 Monthly Revenue Trend

Results

To evaluate business growth over time, monthly revenue was calculated by aggregating sales values for each month between October 2016 and August 2018.

The results show a strong upward trend in revenue throughout the analysis period. Monthly revenue increased from approximately **49.5 thousand BRL** in October 2016 to more than **854.7 thousand BRL** in August 2018.

Revenue growth accelerated significantly during 2017, reaching a peak of approximately **1.01 million BRL** in November 2017. Throughout 2018, monthly revenue remained consistently high, with several months generating nearly **1 million BRL** in sales.

Business Insight

The steady increase in monthly revenue indicates strong business growth and increasing customer activity on the Olist platform. The significant revenue peak observed in November 2017 may be associated with seasonal shopping events and promotional campaigns, highlighting the importance of strategic marketing initiatives.

The sustained high revenue levels during 2018 suggest that Olist successfully expanded its marketplace operations and maintained customer demand over time.

Figure 1. Monthly Revenue Trend (October 2016 – August 2018).



4.3 Monthly Order Volume Growth

Results

To evaluate business growth, the number of unique orders was calculated for each month throughout the analysis period.

The results show a strong increase in monthly order volume over time. Monthly orders grew from **324 orders** in October 2016 to **6,512 orders** in August 2018, demonstrating substantial business expansion.

The highest monthly order volume was recorded in **November 2017**, with **7,544 orders**, representing a monthly growth rate of approximately **62.9%** compared with October 2017. Although monthly fluctuations were observed after this peak, order volume remained consistently high throughout 2018.

Business Insight

The steady increase in monthly order volume indicates continuous growth in customer activity and marketplace adoption. The significant peak in November 2017 is likely associated with seasonal shopping events or promotional campaigns, while the stable order volume during 2018 suggests sustained business performance and customer demand.

Monitoring monthly order growth enables the company to identify seasonal trends, forecast future demand, and improve inventory and logistics planning.

Figure 2. Monthly Order Volume Growth.



4.4 Average Order Value (AOV)

Results

Average Order Value (AOV) was calculated by dividing the total product revenue by the total number of unique orders.

The analysis showed that the **Average Order Value (AOV)** was:

137.75 BRL per order

This indicates that, on average, customers spent approximately **137.75 BRL** on products in each order.

Business Insight

The Average Order Value provides valuable insight into customer purchasing behavior and spending patterns. An AOV of **137.75 BRL** suggests a moderate average spending level per transaction.

Increasing AOV can contribute directly to revenue growth without necessarily increasing the number of orders. Strategies such as product bundling, cross-selling, upselling, and free shipping thresholds can encourage customers to spend more per purchase.

4.5 Revenue by Product Category

Results

To identify the product categories generating the highest revenue, total sales were aggregated for each product category.

The analysis revealed that revenue was concentrated in a small number of categories. The top revenue-generating categories were:

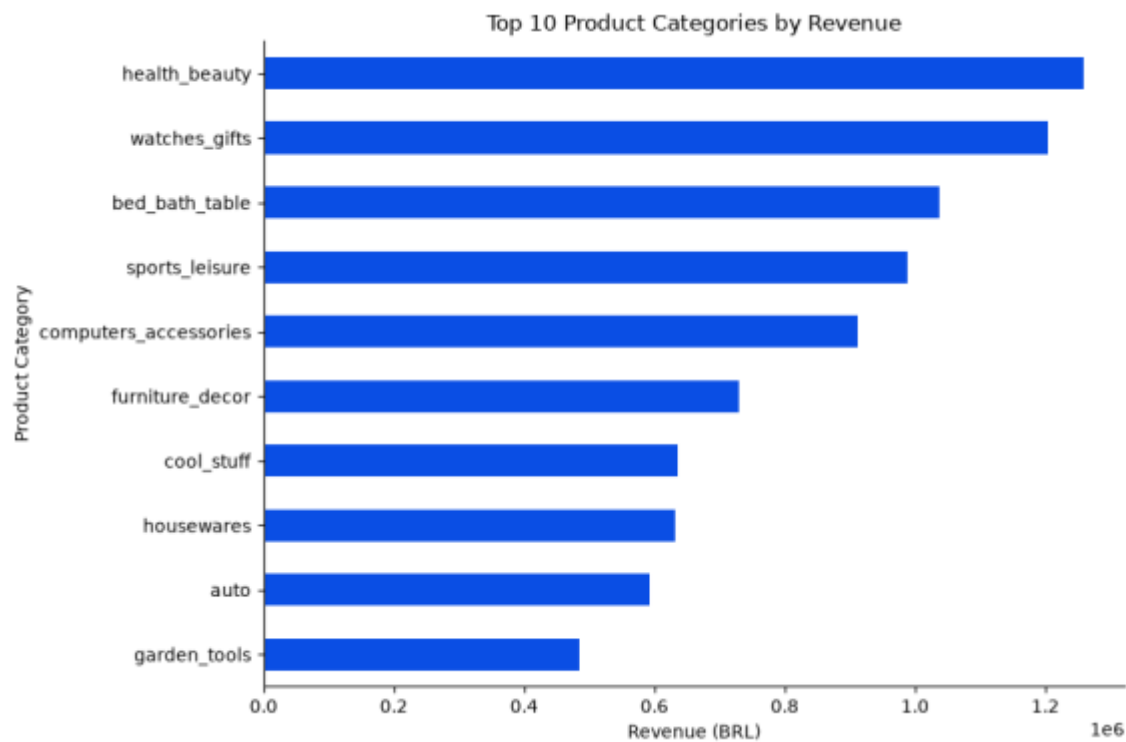
Among all categories, **Health & Beauty** generated the highest revenue, reaching approximately **1.26 million BRL**, followed by **Watches & Gifts** and **Bed, Bath & Table**.

Business Insight

The results indicate that a relatively small number of product categories contribute a significant share of Olist's total revenue. These high-performing categories represent important business assets and should remain a priority for inventory planning, marketing campaigns, and promotional activities.

At the same time, lower-performing categories should be evaluated to identify opportunities for sales improvement, product expansion, or targeted marketing strategies.

Figure 3. Top 10 Product Categories by Revenue.



4.6 Freight Cost Analysis

Results

To better understand shipping costs, the relationship between delivery distance and freight value was analyzed. The Pearson correlation coefficient between **distance** and **freight value** was **0.392**, indicating a moderate positive relationship.

This suggests that freight charges generally increase as the delivery distance becomes longer, although distance alone does not fully explain shipping costs.

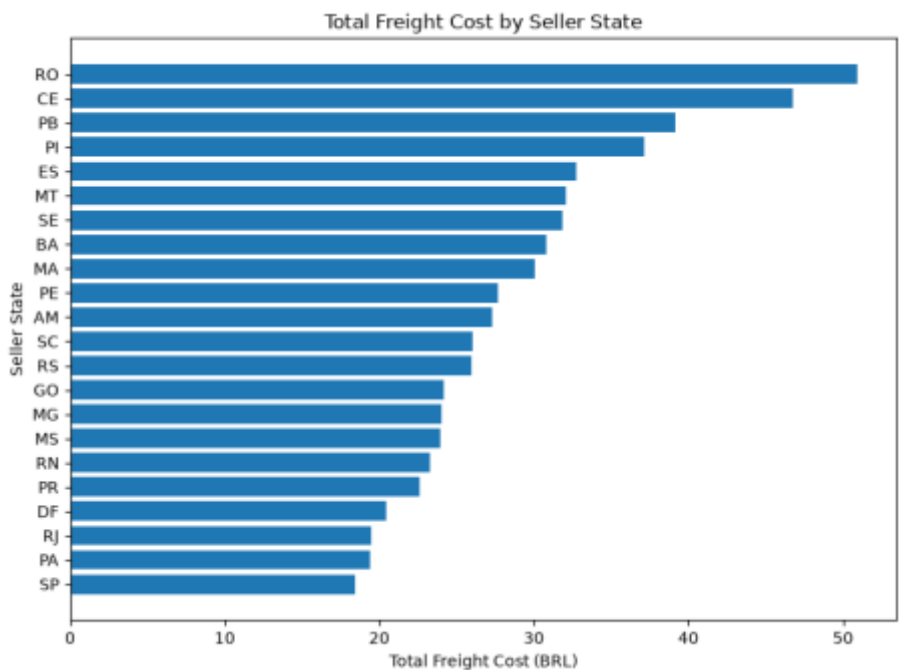
The analysis also compared the average freight cost across seller states. Sellers located in **Rondônia (RO)** had the highest average freight cost (**50.91 BRL**), followed by **Ceará (CE)** (**46.74 BRL**) and **Paraíba (PB)** (**39.16 BRL**). In contrast, **São Paulo (SP)** had the lowest average freight cost among the major seller states, with an average of **18.42 BRL**.

Business Insight

The results indicate that freight cost is influenced by geographical factors, with longer shipping distances generally resulting in higher transportation costs. However, the moderate correlation suggests that other factors, such as package size, product weight, logistics partners, and regional infrastructure, also contribute to freight pricing.

The variation in freight costs across seller states highlights potential regional differences in logistics efficiency and transportation expenses. Understanding these differences can help optimize shipping strategies, improve pricing policies, and reduce operational costs.

Figure 4. Average Freight Cost by Seller State.



4.7 Freight Cost vs. Order Value

Results

To investigate whether higher-value orders are associated with higher shipping costs, the relationship between **order value** and **freight cost** was analyzed.

The Pearson correlation coefficient between the two variables was **0.411**, indicating a **moderate positive correlation**. This suggests that orders with higher values generally tend to have higher freight costs.

The scatter plot shows that most orders are concentrated in the lower order value range with relatively low freight costs. A small number of high-value orders appear as outliers with considerably higher shipping costs.

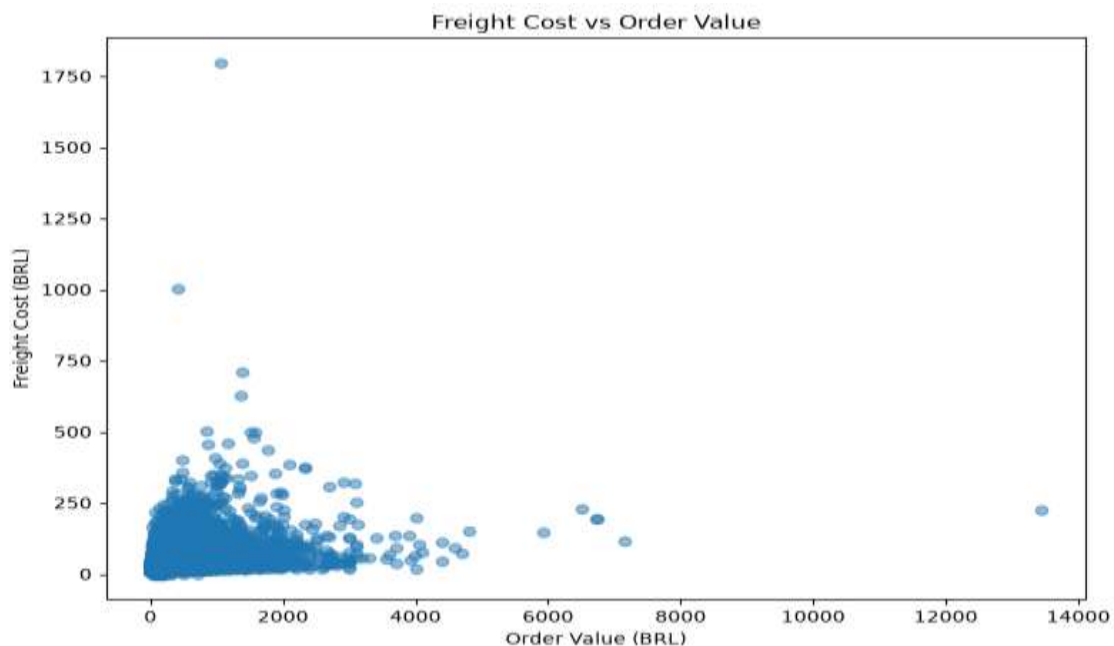
Business Insight

The moderate positive relationship indicates that freight cost tends to increase as order value increases. This may be explained by the fact that higher-value orders often include larger, heavier, or multiple products, leading to higher transportation costs.

However, the correlation is not strong, suggesting that freight cost is also influenced by other factors such as shipping distance, package dimensions, product weight, and logistics operations.

These findings can help Olist optimize shipping strategies, evaluate pricing policies, and design promotional programs such as free shipping thresholds while maintaining operational efficiency.

Figure 5. Relationship Between Order Value and Freight Cost.



4.8 Business Insights

The revenue analysis provides several important insights into Olist's business performance:

- Olist generated a total **Gross Merchandise Value (GMV)** of **13.59 million BRL**, indicating strong overall marketplace performance.
- Monthly revenue showed a clear upward trend throughout the analysis period, demonstrating continuous business growth.
- Order volume increased significantly over time, reflecting a growing customer base and higher marketplace activity.
- The **Average Order Value (AOV)** was **137.75 BRL**, suggesting a stable average customer spending level per transaction.
- Revenue was concentrated in a limited number of product categories, with **Health & Beauty**, **Watches & Gifts**, and **Bed, Bath & Table** generating the highest sales.
- Freight cost showed a moderate positive relationship with both delivery distance and order value, indicating that shipping charges are influenced by multiple operational factors rather than a single variable.

Overall, the results indicate that Olist experienced steady business growth supported by increasing sales, expanding customer demand, and strong performance across its leading product categories.

5. Key Findings

The analysis revealed several important findings regarding Olist's revenue and business growth:

- Olist generated a total **Gross Merchandise Value (GMV)** of **13.59 million BRL** during the analysis period.
- Monthly revenue showed a consistent upward trend, indicating continuous business growth.
- Monthly order volume increased significantly, reflecting an expanding customer base and higher marketplace activity.
- The **Average Order Value (AOV)** was **137.75 BRL**, representing a stable average customer spending level.
- The **Health & Beauty**, **Watches & Gifts**, and **Bed, Bath & Table** categories generated the highest revenues among all product categories.
- Freight cost showed a moderate positive relationship with both delivery distance and order value, suggesting that shipping costs are influenced by multiple operational factors.

6. Recommendations

Based on the analysis, the following recommendations are proposed:

- Continue investing in high-performing product categories to maximize revenue growth.
- Develop targeted marketing campaigns for lower-performing categories to improve sales performance.
- Increase the Average Order Value (AOV) through product bundles, cross-selling, and upselling strategies.
- Optimize freight pricing by considering shipping distance, product characteristics, and logistics efficiency.
- Monitor monthly revenue and order volume regularly to identify seasonal trends and support business planning.
- Evaluate opportunities to improve logistics operations while maintaining competitive shipping costs.

7. Conclusion

The analysis demonstrates that Olist experienced strong business growth during the study period. Revenue and order volume increased steadily, indicating expanding customer demand and marketplace activity.

The results also highlight the importance of leading product categories in driving revenue and show that freight costs are moderately influenced by both delivery distance and order value.

Overall, the findings provide valuable insights that can support strategic decision-making, improve operational efficiency, and contribute to sustainable business growth.

8. Appendix

Business Indicators Used

Gross Merchandise Value (GMV)

Total value of all products sold.

Formula:

GMV = Sum of Product Prices

Average Order Value (AOV)

Average amount spent per order.

Formula:

$AOV = \text{Total Revenue} \div \text{Number of Unique Orders}$

Monthly Revenue

Total sales revenue generated each month.

Monthly Order Volume

Number of unique orders placed each month.

Category Revenue

Total revenue generated by each product category.

Freight Cost

Shipping fee charged for delivering an order.

Order Value

Total monetary value of an individual order.

Pearson Correlation Coefficient

A statistical measure used to evaluate the strength and direction of the relationship between two numerical variables.